Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Analysis
Maximum Marks	2 Marks

Define Your Technology Stack

The project uses a CSV dataset for source data, Tableau for data exploration and visualization, Tableau Public for sharing the dashboard and Story, and a small Flask-based website for presenting project information. The stack below reflects the tools used in the project; Snowflake is not listed because it was not part of the described implementation.

S. No.	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language used for the project website and supporting logic.	Python	Python runs the Flask web application and supports the site's server-side routing.
2	Database / Data Warehouse	Structured source data used for the analysis.	CSV file (global_fuel_prices_Dataset.csv)	The project dataset is stored as a CSV and connected to Tableau; no separate database or warehouse is identified as used.
3	Data Ingestion	Loading the source data into the analytics environment.	Tableau data connection to CSV	Connects the dataset to Tableau so its fields can be used in worksheets, dashboards and the Story.
4	Data Transformation and Business Logic	Preparation and analytical calculations applied to the data.	Tableau data model, field types, aggregations and calculated measures where applicable	Supports comparisons of fuel prices, tax percentage, subsidy level, region, income level and Brent crude prices.
5	Data Visualization / Analytics	Charts, comparisons, dashboards and analytical storytelling.	Tableau Desktop / Tableau Public	Used to create the country comparisons, fuel-price trend charts, tax and subsidy analysis, dashboard and Story.
6	Front-End / User Interface	Web interface for displaying project information.	HTML, CSS, Bootstrap and Flask templates	Used by the project website to present project details and provide a web-based entry point to project materials.
7	Publishing / Sharing	Online access to the completed analytical output.	Tableau Public; GitHub repository	Tableau Public shares the dashboard and Story; GitHub provides repository-based project files where committed.
8	Development Environment	Tools used to edit and maintain project files.	Code editor and web browser	Used to edit website files, inspect the published dashboard and verify project presentation.

Data flow summary: CSV dataset → Tableau data connection → field preparation and analysis → worksheets → dashboard and Story → Tableau Public. The Flask website is a separate presentation layer for project information.